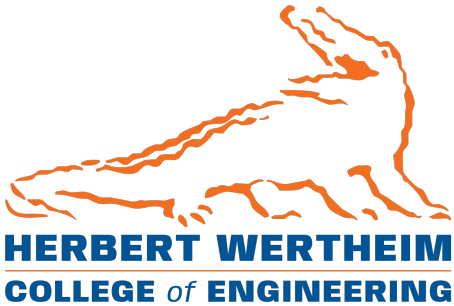
# M3: Alpha Build

## EEL4908 CpE Design 2

Hardened RISC-V Exam Computer

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# 1 Alpha objective

The alpha goal is **Safe Exam Browser running on the Debian board**: a student-facing locked browser on the RISC-V hardware.

Repository: `https://github.com/cannon-spencer/riscv-exam-computer`

CI in `reptilian-riscv` builds the Debian image. Scripts here flash that image and install on-device software (`flash-os.sh`, `install-software.sh`). A board can boot Debian, join the network, and run `seb-agent`, which POSTs a heartbeat to a listener.

# 2 Vertical features

| Element | Alpha |
|---|---|
| External interface | Debian session on the Orange Pi RV (login, network). SEB is the locked student browser on that session. |
| Persistent state | Flashed OS image; `seb-agent` env (`CONTROL_URL`) written by the install script. |
| Internal systems | CI Debian build; `flash-os.sh` and `install-software.sh`; `seb-agent` heartbeat. |

*Table 1: Vertical features in the alpha build.*

# 3 Specification mapping

| Criterion | Alpha |
|---|---|
| Usability | Dedicated Debian board: power on, log in, network. SEB is the exam control surface on that session. |
| Interface | Debian desktop; `seb-agent` heartbeat over HTTP; image and agent env as persistent state. |
| Navigation | Repeatable path: flash, boot, UART or SSH, Wi-Fi via `nmtui`. Scripts wrap flash and software install. |
| Perception | Login and desktop on HDMI; serial/SSH output; `seb-agent` logs each heartbeat. |
| Responsiveness | `seb-agent` yields between POSTs (`sleep` 5 s) rather than busy-waiting on I/O. |
| Robustness | CI image, flash, and boot complete under normal lab use (UART, SSH, outbound IP). |
| Consistency | The same published image and scripts produce a booting Debian board and a heartbeat POST. |
| Aesthetic rigor | Debian session is the on-device surface; SEB supplies the exam chrome. |

*Table 2: M3 specification mapping.*

# 4 Work completed

- CI-built Debian image for the Orange Pi RV; `flash-os.sh` writes it to SD.
- Board boots, UART, Wi-Fi, SSH, outbound IP.
- `install-software.sh` deploys `seb-agent`. A heartbeat POST reaches a LAN listener.
- `seb-linux` is in-tree as a submodule. Prior compile issues on riscv64 (WebKitGTK path) still have to be cleared on this image so SEB can launch.

## 5 Known bugs and limitations

SEB is not yet auto-installed or launched on the image. Agent start/stop and config, instructor UI, Canvas exam flow, and kernel hardening are later work.

| ID | Description | Status |
|---|---|---|
| BUG-01 | Qt6 `signals` vs. GLib/GTK `signals` in the `seb-linux` WebKitGTK backend. | Reported upstream; verify on this image |
| BUG-02 | Spell-check language pointer type in `webkitgtk_profile.cpp`. | Reported upstream; verify on this image |
| BLK-02 | No riscv64 Qt6 WebEngine; WebKitGTK fallback is the board browser path. | Constraint |

*Table 3: Known bugs.*